Unit 3 2023.pdf

Examiner only

4. (a) Explain what is meant by specific heat capacity. [2]

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(b) A volume of $1.2 \times 10^{-3}\,m^3$ of water is heated in a 3 kW kettle.

Specific heat capacity of water (and tea) = $4200\,J\,kg^{-1}\,K^{-1}$

Specific heat capacity of milk = $3900\,J\,kg^{-1}\,K^{-1}$

Density of water = $1.00 \times 10^3\,kg\,m^{-3}$

Density of milk = $1.03 \times 10^3\,kg\,m^{-3}$

(i) Determine the time it takes to increase the temperature of the water from 18 °C to 100 °C. [3]

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(ii) A volume of $0.25 \times 10^{-3}\,m^3$ of the heated water is used to make tea and this is poured into a thermos flask. The temperature of the tea when in the flask is 95 °C. Milk at 5 °C is then poured into the flask. Determine whether $3.6 \times 10^{-5}\,m^3$ of milk will lower the temperature of the mixture to the required temperature of 84 °C. [4]

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